

Appendix A. Supplementary Material

Seaweed-derived biochar with multiple active sites as a heterogeneous catalyst for converting macroalgae into acid-free biooil containing abundant ester and sugar substances

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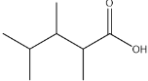
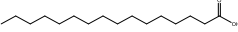
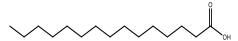
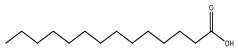
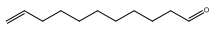
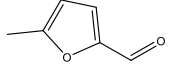
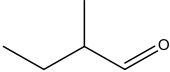
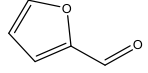
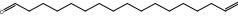
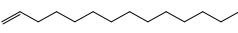
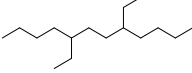
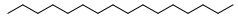
² School of Energy and Power Engineering, Jiangsu University, Zhenjiang 212013, China.

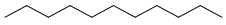
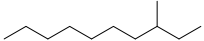
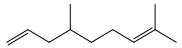
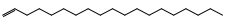
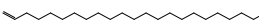
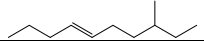
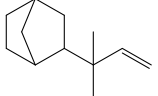
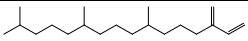
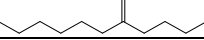
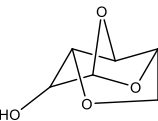
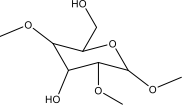
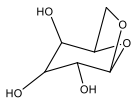
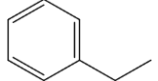
³ State Key Laboratory of Coal Combustion, School of Energy and Power Engineering, Huazhong University of Science and Technology, Wuhan 430074, China

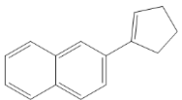
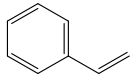
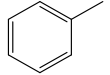
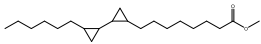
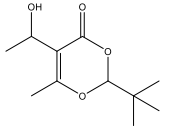
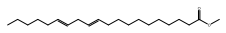
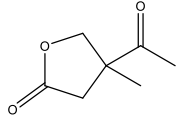
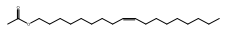
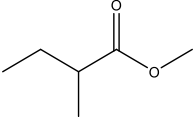
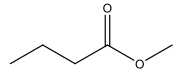
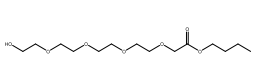
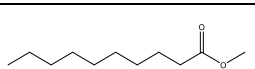
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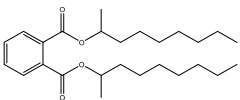
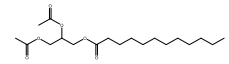
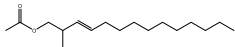
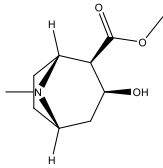
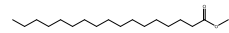
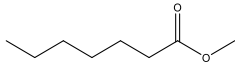
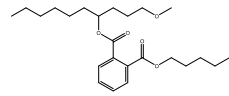
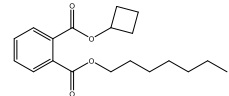
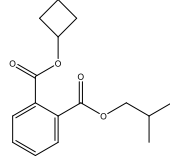
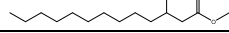
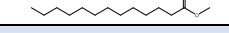
Table S1 Relative contents of chemical compounds in biooils by GC-MS analysis.

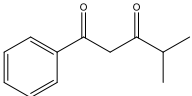
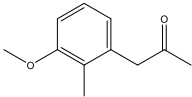
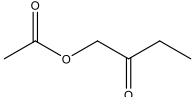
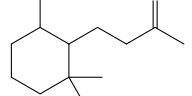
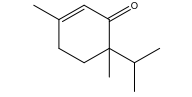
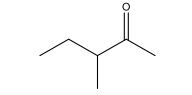
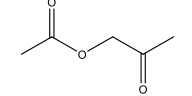
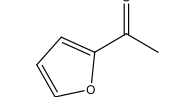
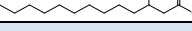
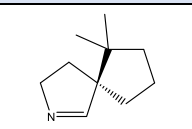
Table S1 Relative contents of chemical compounds in biooils by GC-MS analysis.

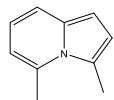
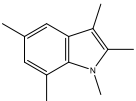
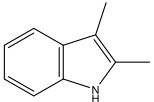
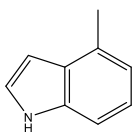
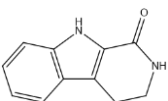
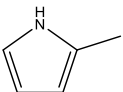
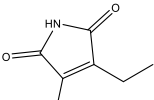
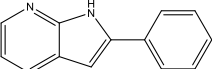
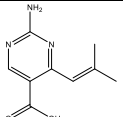
Hit name	Structure	Formula	Relative content (%)						
			Control	Catalytic pyrolysis					
			EN	BC-600	BC-700	BC-800	Cu-BC-800	Ce-BC-800	Fe-BC-800
<i>Carboxylic acids</i>			32.54	2.11	0.00	0.00	0.00	0.00	0.00
2,3,4-Trimethylpentanoic acid		C8H16O2		0.888					
n-Hexadecanoic acid		C16H32O2	30.17						
Pentadecanoic acid		C15H30O2	2.37						
Tetradecanoic acid		C14H28O2		1.221					
<i>Aldehydes</i>			7.66	0.00	0.00	2.48	0.00	0.00	2.79
10-Undecenal		C11H20O							2.789
2-Furancarboxaldehyde, 5-methyl-		C6H6O2	6.03						
Butanal, 2-methyl-		C5H10O	0.5						
Furfural		C5H4O2	1.13						
16-Heptadecenal		C17H32O				2.481			
<i>Aliphatic hydrocarbons</i>			6.45	7.75	4.83	11.12	9.64	6.48	6.47
1-Tetradecene		C14H28						1.035	
Dodecane, 5,8-diethyl-		C16H34		2.633		3.47	3.208	2.45	
Hexadecane		C16H34		1.679				2.996	

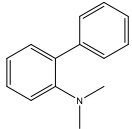
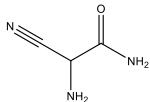
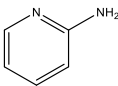
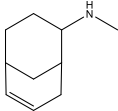
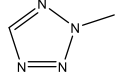
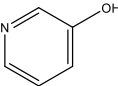
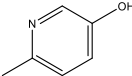
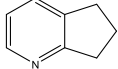
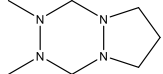
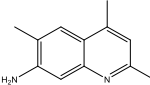
Undecane		C ₁₁ H ₂₄				2.216	6.427		4.127
Decane, 3-methyl-		C ₁₁ H ₂₄			2.337				
1,7-Nonadiene, 4,8-dimethyl-		C ₁₁ H ₂₀							2.346
11-Tricosene		C ₂₃ H ₄₆				1.193			
1-Nonadecene		C ₁₉ H ₃₈		0.627					
1-Tetracosene		C ₂₄ H ₄₈	1.29						
4-Decene, 8-methyl-, (E)-		C ₁₁ H ₂₂	1.06						
Bicyclo[2.2.1]heptane, 2-chloro-1,3,3-trimethyl-, endo-		C ₁₂ H ₂₀			2.494				
Neophytadiene		C ₂₀ H ₃₈	4.1						
Undecane, 5-methylene-		C ₁₂ H ₂₄		0.458					
Hexa-1,3,5-triyne		C ₆ H ₂		2.353		4.236			
<i>Sugars</i>			4.12	13.36	24.06	20.53	17.32	16.75	0.00
1,4:3,6-Dianhydro-.alpha.-d-glucopyranose		C ₆ H ₈ O ₄	2.05	1.333				2.425	
Methyl-2,4-di-O-methyl- α -d-glucopyranoside		C ₉ H ₁₈ O ₆		12.03	24.057	20.526	17.321	14.328	
β -D-Glucopyranose, 1,6-anhydro-		C ₆ H ₁₀ O ₅	2.07						
<i>Aromatic hydrocarbons</i>			3.78	0.00	0.00	0.00	3.54	0.00	0.00
Ethylbenzene		C ₈ H ₁₀	0.55						

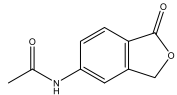
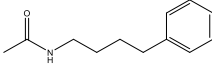
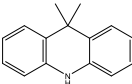
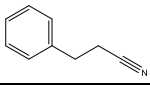
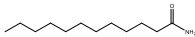
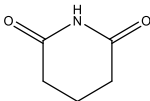
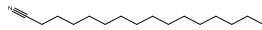
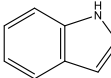
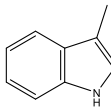
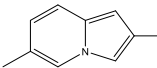
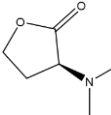
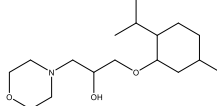
Naphthalene, 2-(1-cyclopenten-1-yl)-		C16H16			6.439				
Styrene		C8H8	0.73				1.139		
Toluene		C7H8	3.05				2.397		
<i>Esters</i>			2.22	8.93	33.60	20.17	35.77	33.55	32.49
[1,1'-Bicyclopropyl]-2-octanoic acid, 2'-hexyl-, methyl ester		C21H38O2							2.138
1,3-Dioxacyclohex-4-en-6-one, 2-t-butyl-5-(1-hydroxyethyl)-4-methyl-		C11H18O4			4.418				
11,14-Eicosadienoic acid, methyl ester		C21H38O2				2.323	4.601	3.233	
3-ACETYL-3-METHYL-.gamma.-BUTYROLACTONE		C7H10O3	0.23						
9-Octadecen-1-ol, acetate, (Z)-		C20H38O2	0.78			1.61			
Butanoic acid, 2-methyl-, methyl ester		C6H12O2					1.248		
Butanoic acid, methyl ester		C5H10O2				1.914	5.035	2.202	2.901
Butyl 14-hydroxy-3,6,9,12-tetraoxatetradecan-1-oate		C14H28O7						3.819	2.406
Decanoic acid, methyl ester		C11H22O2						13.351	10.966

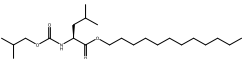
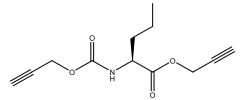
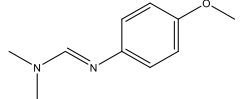
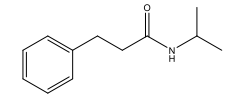
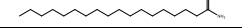
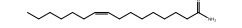
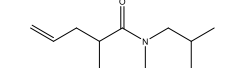
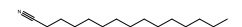
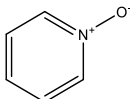
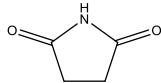
Di-2-nonyl phthalate		C26H42O		1.438		2.778			
Dodecanoic acid, 2,3-bis(acetyloxy)propyl ester		C19H34O6							3.146
E-2-Methyl-3-tetradecen-1-ol acetate		C17H32O2						2.75	
Ecgonine methyl ester		C10H17NO3					4.221		
Heptadecanoic acid, methyl ester		C18H36O2		5.191					
Heptanoic acid, methyl ester		C8H16O2			4.909				
Phthalic acid, 1-methoxydec-4-yl pentyl ester		C24H38O5			4.74				
Phthalic acid, cyclobutyl heptyl ester		C19H26O4						3.5	
Phthalic acid, cyclobutyl isobutyl ester		C16H20O4	1.21	2.301	5.571	2.646	4.18	4.69	
Tridecanoic acid, 3-methyl-, methyl ester		C15H30O2							10.932
Tridecanoic acid, methyl ester		C14H28O2			13.957	8.901	16.485		
Ketones			4.13	1.01	2.39	2.26	0.00	0.00	6.06

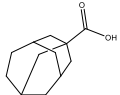
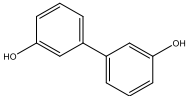
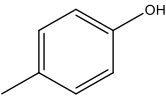
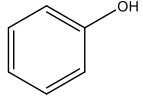
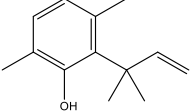
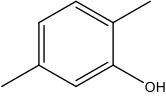
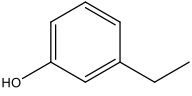
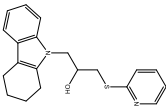
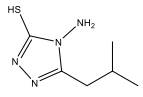
1,3-Pentanedione, 4-methyl-1-phenyl-		C ₁₂ H ₁₄ O ₂			2.386	2.261			
3-Methoxy-2-methylphenylacetone		C ₈ H ₁₀ O ₂							1.553
2-Butanone, 1-(acetyloxy)-		C ₆ H ₁₀ O ₃	1.53						
2-Butanone, 4-(2,2,6-trimethylcyclohexyl)-		C ₁₃ H ₂₄ O							4.503
2-Cyclohexen-1-one, 3,6-dimethyl-6-(1-methylethyl)-		C ₁₁ H ₁₈ O		1.008					
2-Pentanone, 3-methyl-		C ₆ H ₁₂ O	0.61						
2-Propanone, 1-(acetyloxy)-		C ₅ H ₈ O ₃	0.36						
Ethanone, 1-(2-furanyl)-		C ₆ H ₆ O ₂	0.46						
Methyl 3-hydroxytetradecanoate		C ₁₅ H ₃₀ O ₃	1.17						
<i>N-containing compounds</i>			28.67	45.12	28.69	33.86	33.74	38.27	44.78
(S)-6,6-Dimethyl-2-azaspiro[4.4]non-1-ene		C ₁₀ H ₁₇ N							5.29

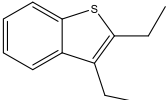
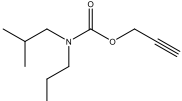
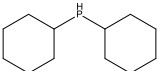
1 Indolizine, 2,8-dimethyl-		C10H11N	0.59						
1H-Indole, 1,2,3,5,7-pentamethyl-c		C13H17N		0.697					
1H-Indole, 2,3-dimethyl-		C10H11N		1.472			3.68	6.661	3.837
1H-Indole, 4-methyl-		C9H9N		3.866		8.569	2.80	10.083	6.11
1H-Pyrido[3,4-b]indol-1-one, 2,3,4,9-tetrahydro-		C11H10N2O			9.781				
1H-Pyrrole, 2-methyl-		C5H7N	0.31						
1H-Pyrrole-2,5-dione, 3-ethyl-4-methyl-		C7H9NO2	0.28						
1H-Pyrrolo[2,3-b]pyridine, 2-phenyl-		C13H10N2				3.525			
2-Amino-4-(2-methylpropenyl)-pyrimidin-5-carboxylic acid		C9H11N3O2			5.566				

2-Aminobiphenyl, N,N-dimethyl-		C14H15N			2.723				
2-Aminocyanoacetamide		C3H5N3O		1.152					
2-Aminopyridine		C5H6N2		1.397					
2-Bicyclo[3.3.1]nonene, 6-methylamino-		C10H17N							2.662
2H-Tetrazole, 2-methyl-		C2H4N4					3.71%		
3-Pyridinol		C5H5NO	11.39						
3-Pyridinol, 6-methyl-		C6H7NO	1.81						
5H-1-Pyridine, 6,7-dihydro-		C10H14N2	1.09	1.404					
6H-Pyrazolo[1,2-a][1,2,4,5]tetrazine, hexahydro-2,3-dimethyl-		C7H16N4							2.917
7-Amino-2,4,6-trimethylquinoline		C12H14N2						4.331	

Acetamide, N-(1,3-dihydro-1-oxo-5-isobenzofuryl)-		C10H9NO3					3.958		
Acetamide, N-(4-phenylbutyl)-		C16H21NO		0.765					
Acridine, 9,10-dihydro-9,9-dimethyl-		C15H15N		2.35				2.047	
Benzenepropanenitrile		C9H9N	0.73	1.145					
Dodecanamide		C12H25NO		7.066					
Glutarimide		C5H7NO2	1.75						
Hexadecanenitrile		C16H31N		5.661		7.031	5.234	4.029	10.532
Indole		C8H7N	2.19	5.494		6.498	0.632	7.022	2.331
Indole, 3-methyl-		C9H9N	1.47		4.371		3.505		
Indolizine, 2,6-dimethyl-		C10H11N		1.709		5.142	4.861	1.881	3.402
L-Homoserine lactone, N,N-dimethyl-		C6H11NO2	0.62						
Propan-2-ol, 1-(2-isopropyl-5-methylcyclohexyloxy)-3-(4-morpholyl)-		C18H35NO2	0.59						

l-Leucine, N-isobutoxycarbonyl-, dodecyl ester		C15H29NO4					5.92		
l-Norvaline, n-propargyloxycarbonyl-, propargyl ester		C12H15NO4		0.679					
Methanimidamide, N'-(4-methoxyphenyl)-N,N-dimethyl-		C10H14N2O							1.413
N-Isopropyl-3-phenylpropanamide		C12H17NO							1.185
Octadecanamide		C18H37NO			6.253	3.099			
Palmitoleamide		C16H31NO		7.111					
Pent-4-enoylamide, 2,N-dimethyl-N-isobutyl-		C15H29NO	1.35						2.795
Pentadecanenitrile		C15H29N	0.53						
Pyridine		C5H5N	0.72						
Pyridine, 1-oxide		C5H5NO		3.153					
Pyrrole		C4H5N	1.35				3.115	2.22	
Succinimide		C4H5NO2	1.9						

Tricyclo[4.3.1.1(3,8)]undecane-1-carboxylic acid		C ₁₁ H ₁₉ N							2.302
<i>Phenols</i>			9.22	18.23	0.00	9.58	0.00	4.94	7.42
[1,1'-Biphenyl]-3,3'-diol		C ₁₂ H ₁₀ O ₂	2.25	6.357	4.53	6.545			1.896
p-Cresol		C ₇ H ₈ O	4.17	9.828	1.06	3.034		3.262	
Phenol		C ₆ H ₆ O	1.8						
Phenol, 2-(1,1-dimethyl-2-propenyl)-3,6-dimethyl-		C ₁₃ H ₁₈ O						1.68	2.528
Phenol, 2,5-dimethyl-		C ₈ H ₁₀ O	0.3	0.549					
Phenol, 3-ethyl-		C ₈ H ₁₀ O	0.7	1.491	0.84				2.994
<i>Others</i>			1.21	3.50	0.00	0.00	0.00	0.00	0.00
1H-Carbazole-9-ethanol, 2,3,4,9-tetrahydro-.alpha.-(2-pyridinylthio)methyl]-		C ₁₉ H ₂₃ N ₃ OS		0.671					
4-Amino-5-(2-methylpropyl)-1,2,4-triazole-3-thiol		C ₆ H ₁₂ N ₄ S		0.52					

Benzo[b]thiophene, 2,3-diethyl-		C ₁₂ H ₁₄ S		1.05					
Carbonic acid, monoamide, N-propyl-N-isobutyl-, propargyl ester		C ₁₃ H ₁₀ Cl ₇ NO 5	0.53	0.743					
Phosphine, dicyclohexyl-		C ₁₂ H ₂₃ P		0.513					
Thiophene, 2,3-dihydro-		C ₄ H ₆ S	0.68						